HANDOVER - BLOCK 2 (12-15 NOV)

Handover to Mona: what is ready and what to do next

From Farida - 12 November 2026

Links

Live tracker (update your tasks here):
https://docs.google.com/spreadsheets/d/1JH4ZOhLBre2fi6JLzDMeKQmHHIarsd1svu3o1-JARoo/edit

Shared Drive folder 3dgs-project:
https://drive.google.com/drive/folders/1VXgL1JKVs8lrfZSx1xRJKzLIpw3f48NM?usp=drive_link

GitHub (notebooks, scripts, LOG.md): https://github.com/Farida-gaber44/sparse-view-3dgs-ablation

1. Where we are

- **Setup and baseline are done.** Full InstantSplat (C3) on garden + bonsai at 3 / 6 / 9 photos is consistent with the published Free360 numbers (PSNR 14.72 / 16.61 / 17.34 vs 14.10 / 15.83 / 17.16).
- **C2 (refinement off) works.** Tested on garden, 3 photos: PSNR 14.56, camera poses unchanged (0.00).
- **Everything is saved on Drive,** so a new Colab session needs no download and no rebuild: setup in about 4 minutes, the AI step (MASt3R) is reused from Drive.
- **An independent audit checked all code and methods.** Its fixes are already in the notebook (details in LOG.md, 12 Nov).
- **Your block:** write the two remaining switches (C4 random points, C1 reference poses), test them, then run the pilot on 2 scenes.

2. One-time setup (10 minutes)

1. Google Drive: open "Shared with me", right-click the folder 3dgs-project -> Organize -> Add shortcut -> My Drive. Colab then finds it at MyDrive/3dgs-project.
2. GitHub: download notebooks/02_experiments.ipynb.
3. Colab: File -> Upload notebook -> choose 02_experiments.ipynb. Then Runtime -> Change runtime type -> T4 GPU.
4. When Colab asks for Google Drive access, choose your account and click Allow.

3. Your steps

Step A - check that your account works (tracker M1.1, ~15 min)

In the Settings cell set:

```
CONFIGS_TO_RUN = ["C2"]
SCENES = ["bonsai"]
VIEWS = [3]
```

Then Runtime -> Run all. Done when Step 5 prints "poses changed by 0.00e+00 -> OK". Write the PSNR in the tracker (M1.1).

Step B - write the C4 switch: random points (tracker M2.1)

- Open the cell "Step 2 - Switches" and write the function use_random_points(...). The notes above the cell say exactly which files to change.
- Same number of points as MASt3R, uniform in the box around (points + camera centres), seed ctx.seed, random colours.

- Set confidence_dsp.npy to a constant (its mean), because random points have no MASt3R confidence.
- Delete the line use_random_points.not_ready = True when the function is written.
- Test: CONFIGS_TO_RUN = ["C4"], SCENES = ["garden"], VIEWS = [3]. Expect a lower PSNR than C3 (14.67) and the switch information (number of points, box) printed in Step 5.

Step C - write the C1 switch: reference poses (tracker M3.1, M3.2)

- Write use_reference_poses(...) in the same cell, following the notes: map the dataset's reference poses INTO the MASt3R frame (keep MASt3R points and scale), write images.txt and images.bin in sparse_<n>/0, and the test-camera start in sparse_<n>/1. Keep the MASt3R focal length.
- Delete the line use_reference_poses.not_ready = True.
- Test on garden, 3 photos. Checks: the training-camera error after the switch should be close to 0, and PSNR should be at least as good as C3 (14.67). Look at a few images in Drive/results/C1/garden/3_views/test_renders.

Step D - pilot on 2 scenes (tracker M4.1, ~4 GPU-hours)

```
CONFIGS_TO_RUN = ["C1", "C2", "C3", "C4"]
SCENES = ["garden", "bonsai"]
VIEWS = [3, 6, 9]
```

- Step 3 (Plan) lists only the runs still missing (C3 and C2 garden 3 are already done) and the time needed.
- Run all configurations of one scene in the same Colab session. If the session stops, just Run all again: finished runs are skipped.
- Step 6 writes the tables (means per configuration and the three effects C1-C2, C3-C4, C3-C2).

Step E - finish the block (15 Nov)

- Update the tracker (M2.1, M3.1, M3.2, M4.1) and write the numbers and any problem in LOG.md.
- Hand-off call with Farida (H2): show the pilot tables; Farida re-runs one C4 run on her account.

4. Rules that keep the results fair

Rule	Why
Do not change RES_DIR, VIEWS rule, iterations or the patches	Every run must use the same protocol
Run all configurations of a scene in the same session	Colab software can change between days
Differences below ~0.3 dB PSNR are noise unless repeated	The same run varies by up to ~0.3 dB
Compare camera-position error (ATE) only at the same photo count	Fewer cameras make ATE look smaller
Never edit files in Drive/results or Drive/init_cache by hand	They are the evidence for the paper

If something fails

Copy the last lines of the error (Step 4 prints them) and send them to Farida with the scene, configuration and photo count.

A failed run does not stop the others; after a fix, Run all again and only the missing runs start.